

Customer Shopping Behavior Analysis

Project Report | Python · SQL · Power BI

1. Project Overview

This portfolio project performs a complete end-to-end data analysis on a **customer shopping behavior dataset** containing **5,050 records and 17 columns**. The project follows a structured data analytics pipeline: Exploratory Data Analysis (EDA) in Python, business problem solving in SQL (PostgreSQL), and an interactive dashboard in Power BI.

5,000 Clean Records	17 Columns	5 Categories	12 SQL Queries
------------------------	---------------	-----------------	-------------------

2. Phase 1 — Python EDA (Exploratory Data Analysis)

2.1 Steps Performed

- Data Import: Loaded CSV using `pd.read_csv()`. Verified shape (5050 rows × 17 cols) using `df.shape`, `df.info()`, `df.describe()`.
- Category Correction: Products were assigned to wrong categories. Fixed using a `correct_mapping` dictionary to map all items to correct categories.
- Null Value Treatment: Size — set 'Not Applicable' (Electronics), 'Free Size' (Accessories), mode (Clothing). Review Rating & Purchase Amount — filled with per-item group mean. Previous Purchases — filled with 0.
- Duplicate Removal: Detected 50 duplicate Customer IDs. Removed using `df.drop_duplicates(subset=['Customer ID'], keep='first')`. Final count: 5,000 records.
- Column Standardization: Converted column names to lowercase snake_case. Renamed 'purchase_amount_(usd)' to 'purchase_amount'.
- Data Export: Loaded clean DataFrame into PostgreSQL using SQLAlchemy — `df.to_sql(name='customer', con=engine, if_exists='replace')`.

2.2 Key EDA Findings

Stage	Total Records	Null Values	Categories	Column Names
Before Cleaning	5,050	2,075 nulls	Category mismatch	Mixed case
After Cleaning	5,000	0 nulls	All 5 correct	snake_case

3. Phase 2 — SQL Business Problem Solving (PostgreSQL)

12 business questions were written and solved using SQL. Each query includes: Business Problem → SQL Query → Data Output → Business Impact.

#	Business Question	Key Result	Business Impact
Q1	Highest Revenue Category	Electronics: \$486,367	Stock & marketing decisions
Q2	Impact of Discounts	Avg +\$37 with discount	Smarter discount strategy
Q3	Revenue by Gender	Male > Female > Other	Targeted campaigns
Q4	High-value Discount Customers	10 sample high spenders found	VIP customer retention
Q5	Top/Bottom Product Ratings	Gloves top, Phone lowest (2.94)	Product quality improvement
Q6	Shipping Type vs Revenue	Express: \$429K, avg \$358	Premium shipping pricing
Q7	Subscriber vs Non-Subscriber	Subscribers spend 63% more	Expand subscription program
Q8	Highest Discount Usage %	Laptop 51.7%, Phone 51.5%	Fix margin leakage
Q9	Customer Segmentation	Loyal 61%, Returning 27%	Personalized marketing
Q10	Top 3 Products per Category	Shirt(315), Shoes(303) lead	Product placement & promo
Q11	Repeat Buyers & Subscription	Only 30% repeat = subscribed	Subscription upsell drive
Q12	Revenue by Age Group	51+ = \$399K (highest)	Age-targeted marketing

4. Phase 3 — Power BI Dashboard

An interactive Power BI dashboard was built using the cleaned dataset exported from Python. The dashboard enables non-technical stakeholders to explore data through visual charts and slicers.

4.1 Dashboard Components

- KPI Cards: Total Revenue, Average Purchase Amount, Total Customers, Average Review Rating
- Bar Chart: Revenue by Product Category — Electronics clearly dominant
- Donut Chart: Revenue split by Gender (Male 39.4%, Female 35.5%, Other 25.3%)
- Column Chart: Revenue by Age Group — 51+ leads with \$399K
- Stacked Bar Chart: Revenue by Shipping Type — Express vs Standard vs Free
- Clustered Bar: Subscription Status vs Average Purchase Amount
- Interactive Slicers: Category, Gender, Discount Applied, Season, Subscription Status

5. Key Insights & Recommendations

Insight	Finding	Recommendation
Electronics is #1	Generates 48% of total revenue (\$486K).	Increase stock, run promotions
Discounts are profitable	Customers with discount spend \$37 more on avg.	Use selective smart discounts
Subscribers are premium	Subscribers spend 63% more than non-subscribers.	Grow subscription base
Express shipping pays off	Express = \$429K vs Free Shipping = \$41K.	Promote express as premium
51+ customers are top	\$399K from 51+ age group — highest of all.	Market to older demographics
Low subscription conversion	Only 30% of loyal buyers are subscribed.	Launch loyalty-to-sub campaign